AIN SHAMS UNIVERSITY-

FACULTY OF ENGINEERING

[MCT232s]

Electronics for Instrumentation

PID control circuit

Project report

Team 27

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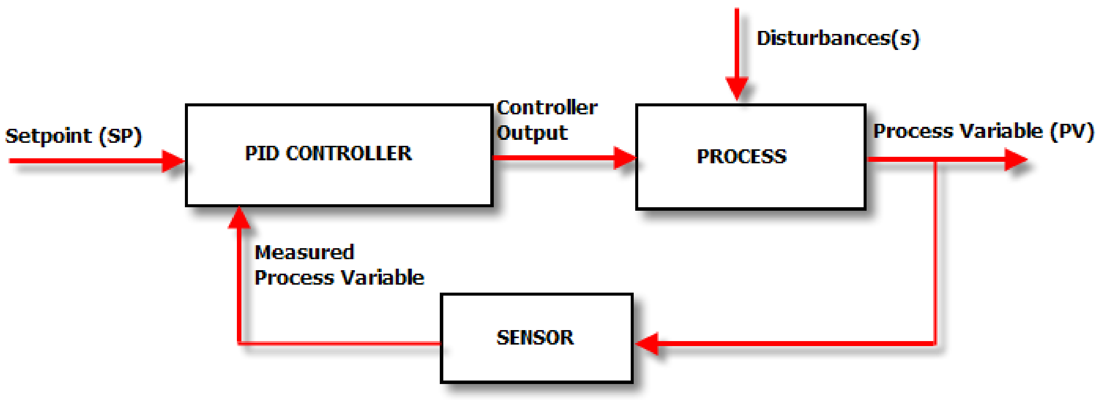
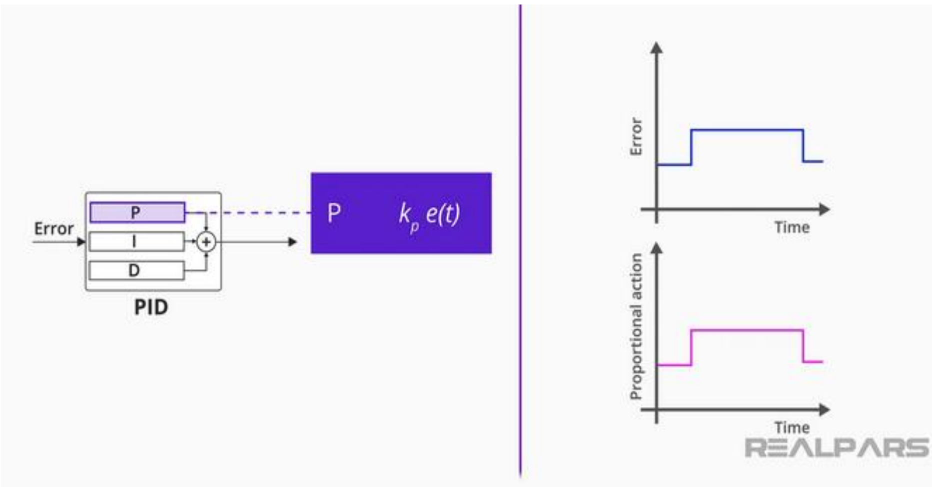
Introduction:

A PID controller, which stands for Proportional-Integral-Derivative controller, is a widely used tool in industrial control systems for regulating various physical quantities like temperature, flow, pressure, and speed. It utilizes a feedback loop to continuously monitor and adjust a process variable to match a desired setpoint.

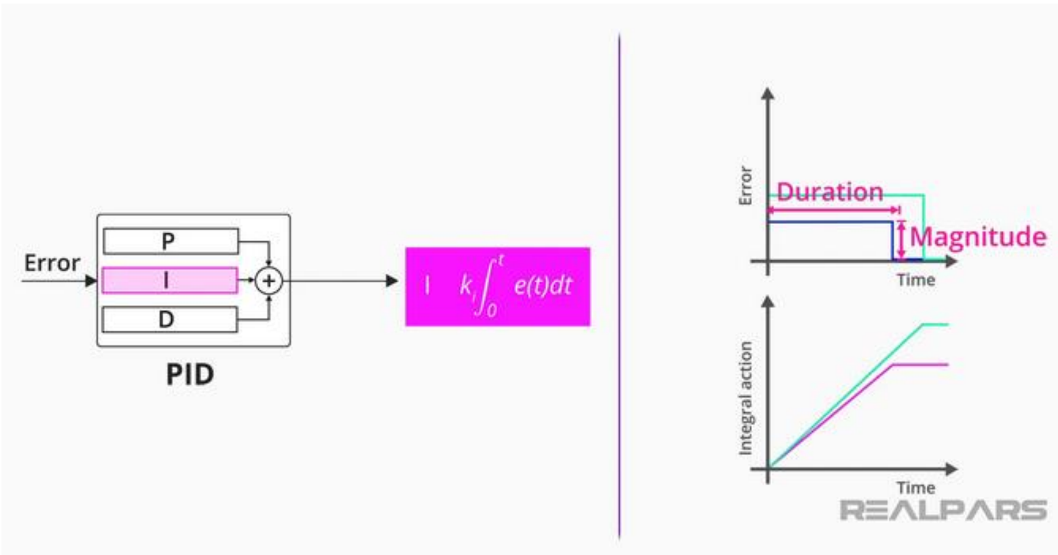


Operation principle:

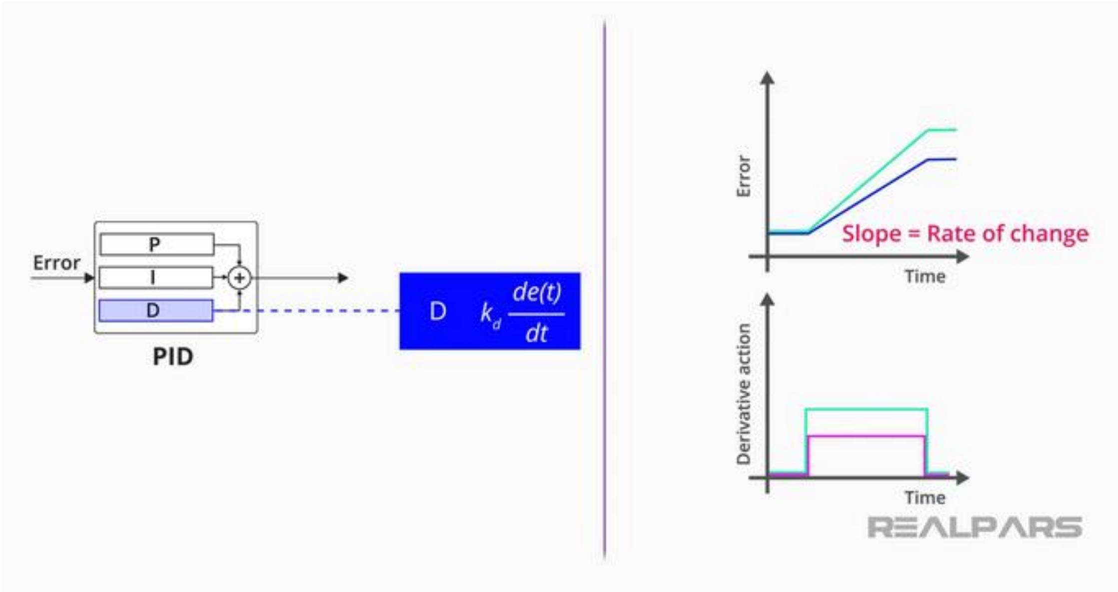
1. **Measures the Error**: The PID controller constantly watches the difference between the setpoint (SP) and the actual measured value of the output (PV), calculating the error.
2. **Threefold Response**: The controller then calculates a response based on three terms:
   - Proportional (P) Term: This term reacts to the current magnitude of the error. A larger error results in a larger response from the controller.



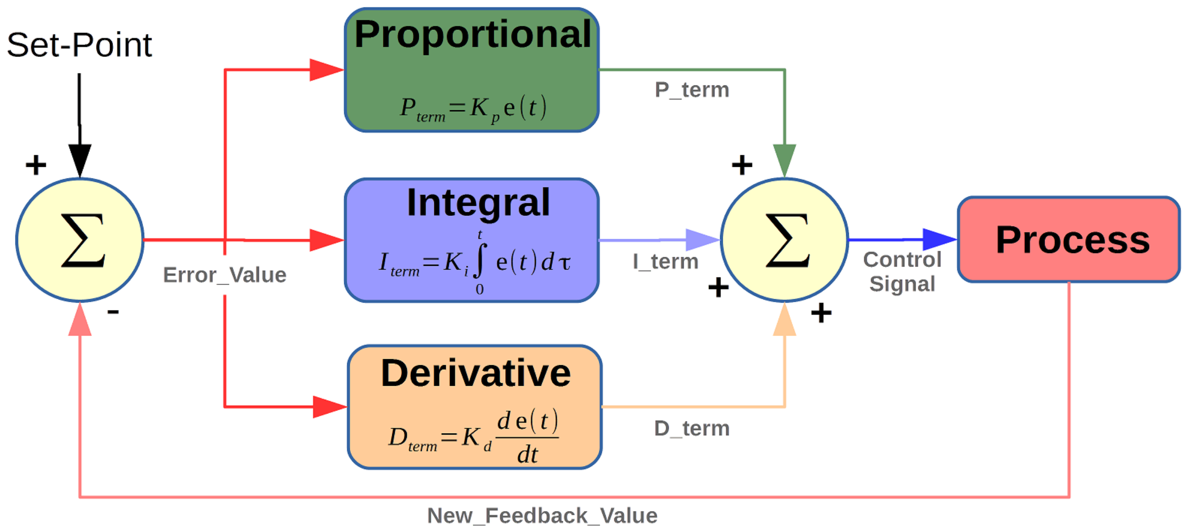
- Integral (I) Term: This term addresses long-term errors by accumulating the error over time. It helps eliminate any steady-state error, where the process variable remains slightly off the setpoint.



- Derivative (D) Term: This term anticipates future errors based on the rate of change of the error signal. It helps the controller react quickly to changes and reduces overshoot, where the process variable briefly goes beyond the desired setpoint.



3. **Output Adjustment**: By summing the proportional, integral, and derivative terms, the PID controller generates a control signal. This signal is sent to a control element, such as a valve or heater, which adjusts the process to minimize the error and bring the process variable closer to the setpoint.



Pros of PID Controllers:

Versatility: They can be applied to a vast range of processes thanks to their adjustable parameters. They can handle variables like temperature, pressure, flow, and speed across various industries.

Effectiveness: The combination of proportional, integral, and derivative actions in a PID controller allows for precise and stable control. The proportional term reacts immediately to errors, the integral term eliminates long-term drifts, and the derivative term anticipates future changes to prevent overshoot.

Simplicity: The core concept behind PID control is relatively easy to understand. While tuning the specific parameters (P, I, D) requires some know-how, the basic principles are accessible.

Cost-effective: PID controllers can be implemented using various technologies, from analog circuits to digital microprocessors. This makes them adaptable and affordable for a wide range of applications.

Cons of PID Controllers:

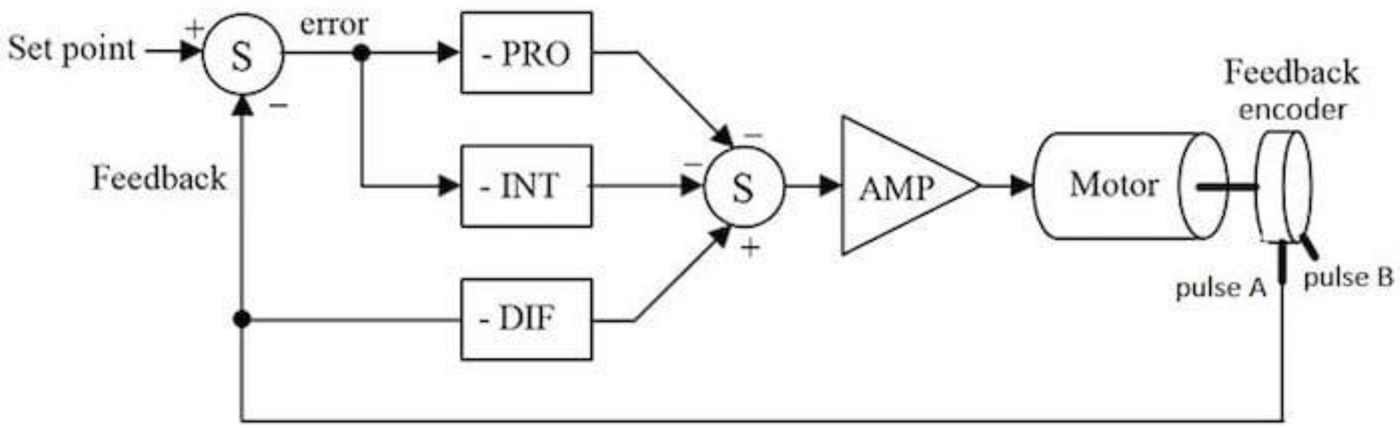
Tuning Time: While the concept is simple, achieving optimal performance from a PID controller requires careful tuning of the proportional, integral, and derivative gains. This tuning process can be time-consuming, especially for complex systems.

Limitations with Non-Linear Systems: PID controllers work best with linear systems where the output has a proportional relationship to the input. In highly non-linear systems, achieving optimal control with a PID controller can be challenging.

Not Ideal for Processes with Large Time Delays: If there's a significant delay between the control action and the observed effect in the process, a PID controller might struggle to maintain stability.

Project objective:

The project objective is to design a PID circuit using op amps to control a motor position-angle when an input set-point is used with minimal error and fastest speed possible.

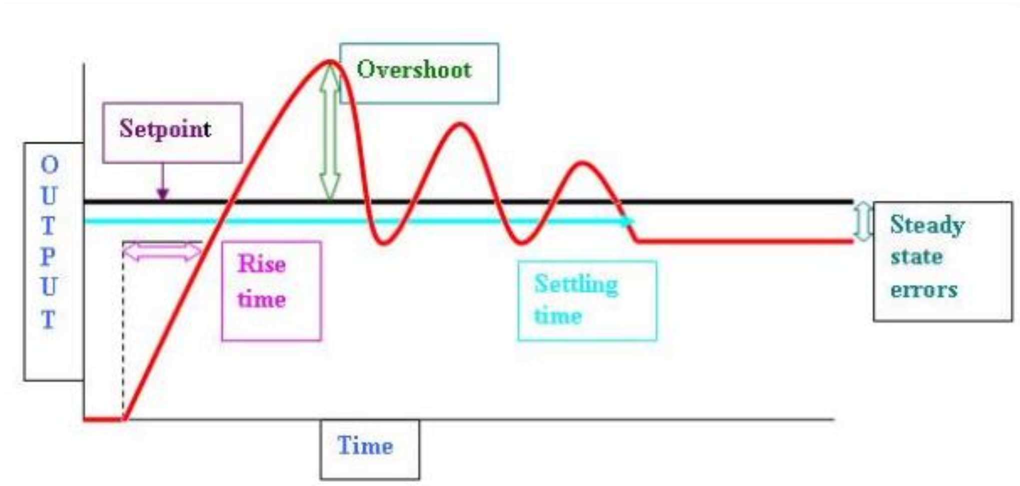


Desired response:

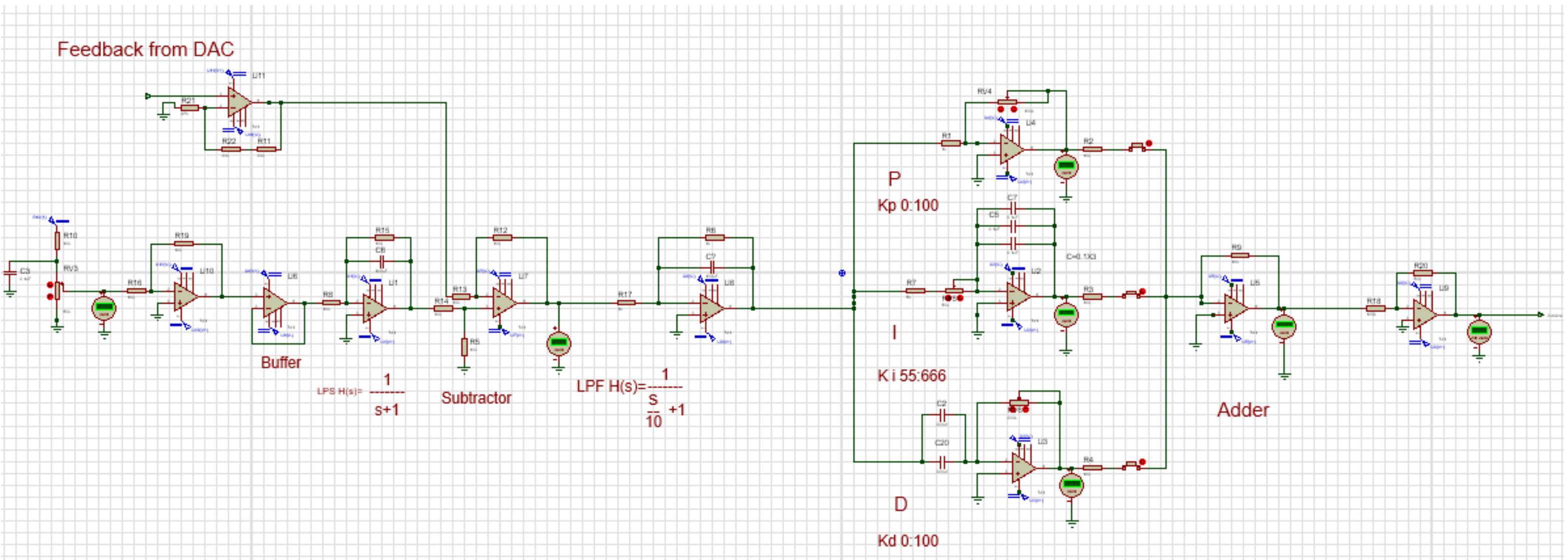
1. **Risetime:** the time the response rise to the set point.
2. **Overshoot:** is the amount the value rise above the set point

3. **Settling time:** the time the response settles to steady state.
4. **steady-state error:** the value that the response stays on away from the setpoint**.**

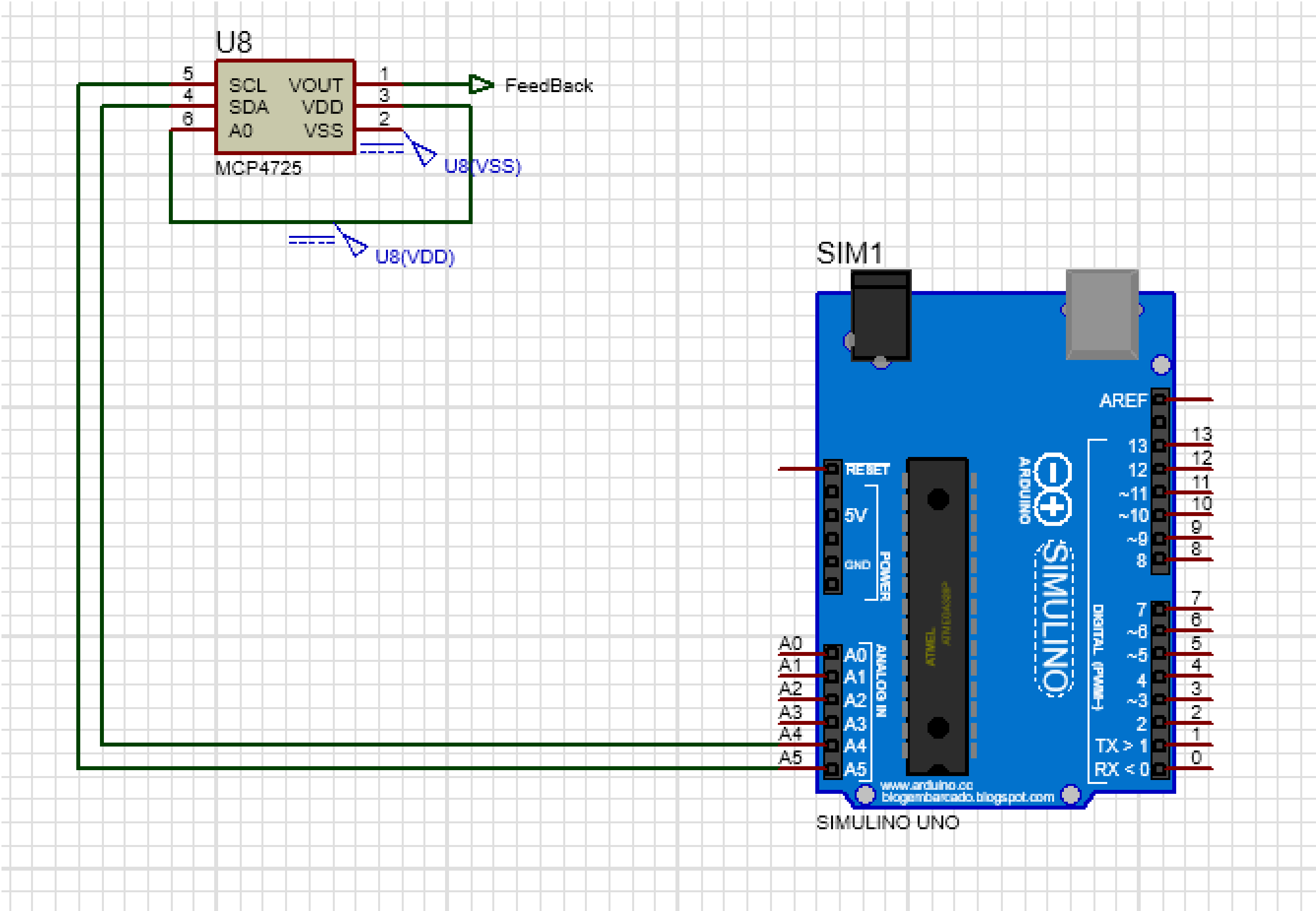
-And they all should be minimised



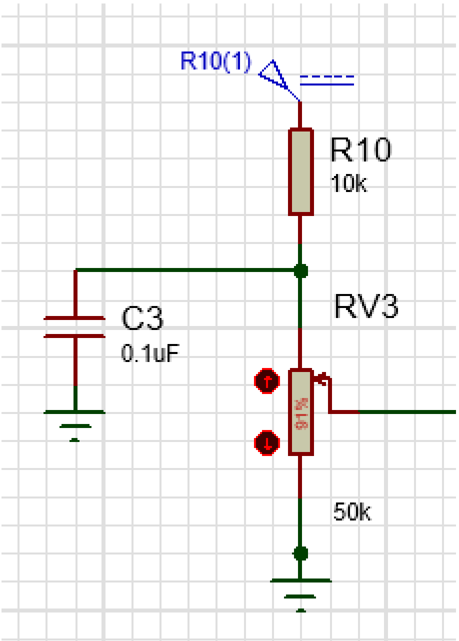
These parameters can be controlled using the value of Kp,Ki,Kd



Circuit:



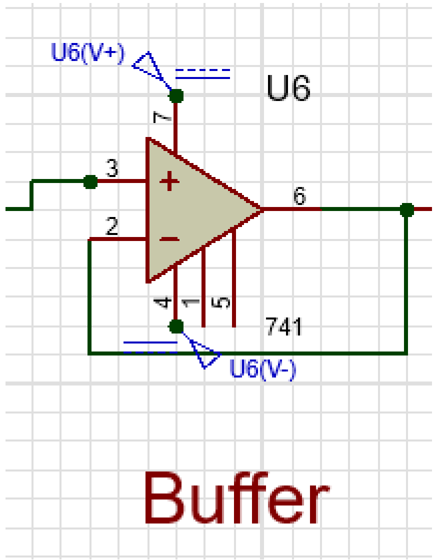
Circuit parts used:



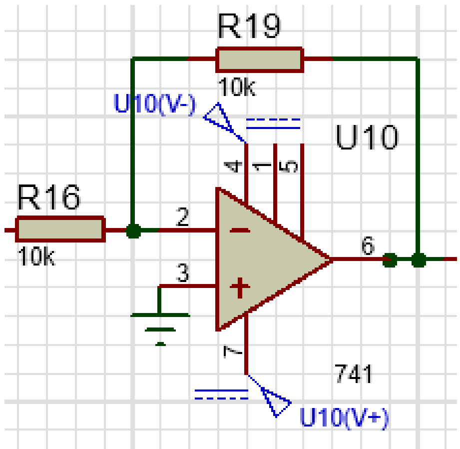
1. input set-point: the potentiometer acts as a voltage divider. As you rotate the knob, changing the voltage output between two terminals. This voltage output then is set to be the desired setpoint range.

This circuit doesn't let the voltage to be 5 v or 0v

2. **buffer**: A buffer is used to isolate the load from the source.



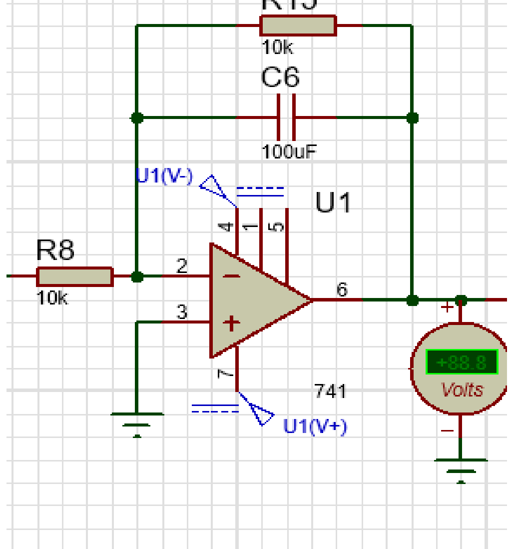
3. **inverting amplifier**: it is used to make the the signal positive with unity gain.



Inverting amp.

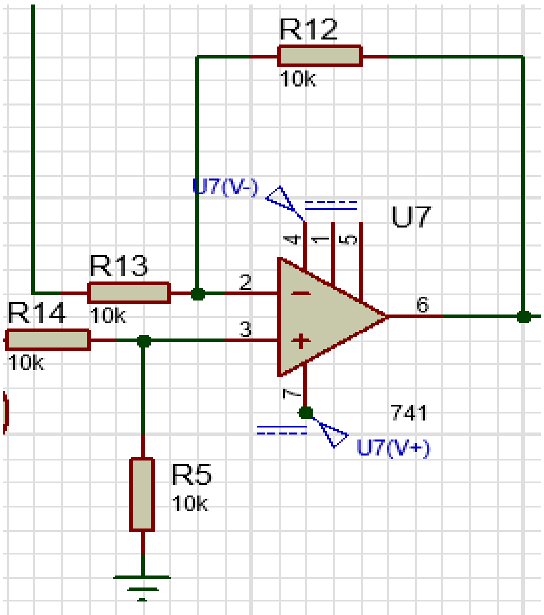
4. **low-pass filter:** a filter is added before the circuit to remove any high frequency noise from the system.

And to rise the voltage smoothly .

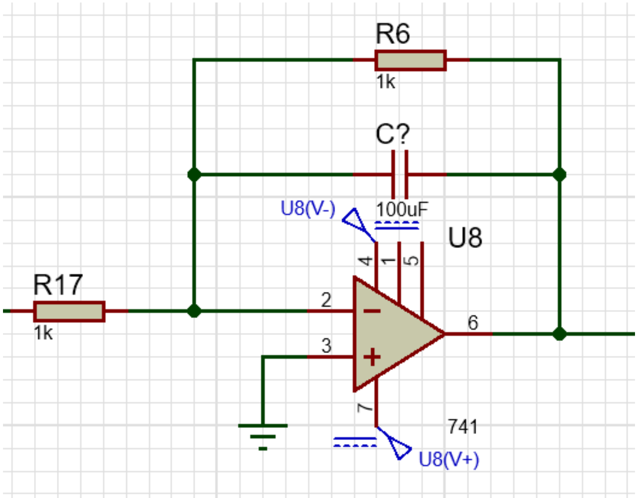


feedback

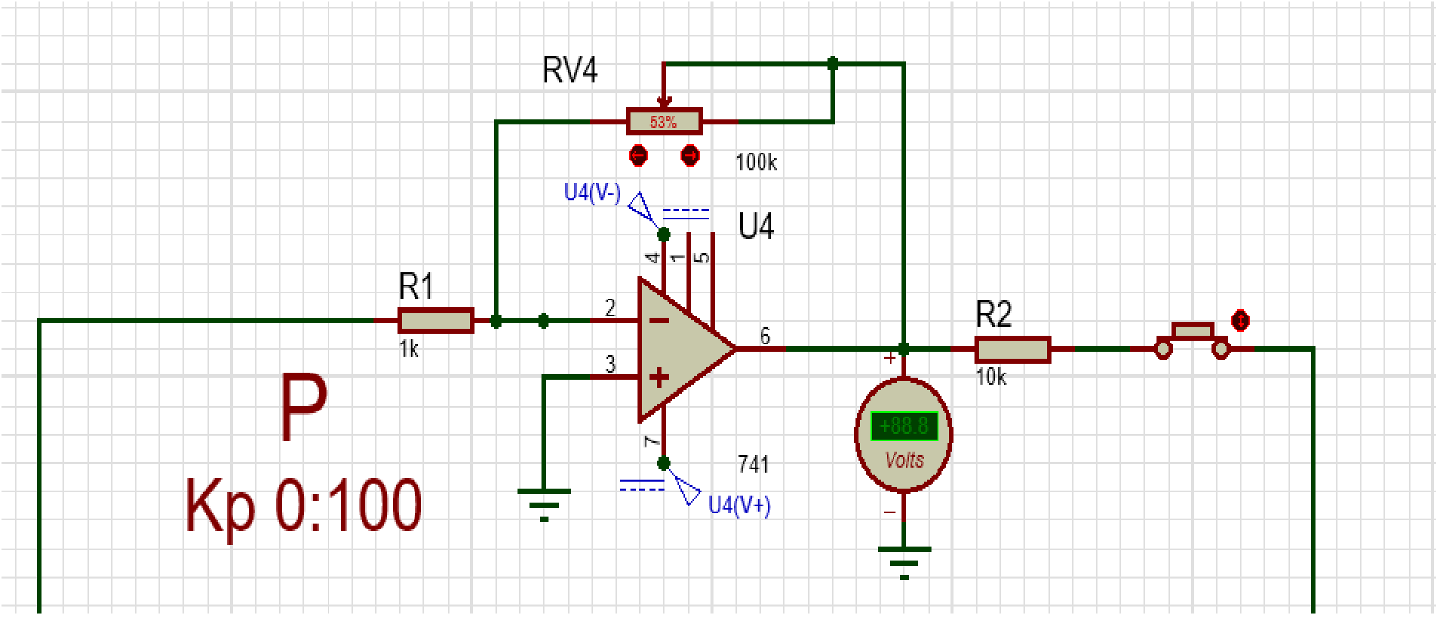
5. **Subtractor:** The subtractor is used as a comparator it compares the feedback with the input signal and check if there is an error.



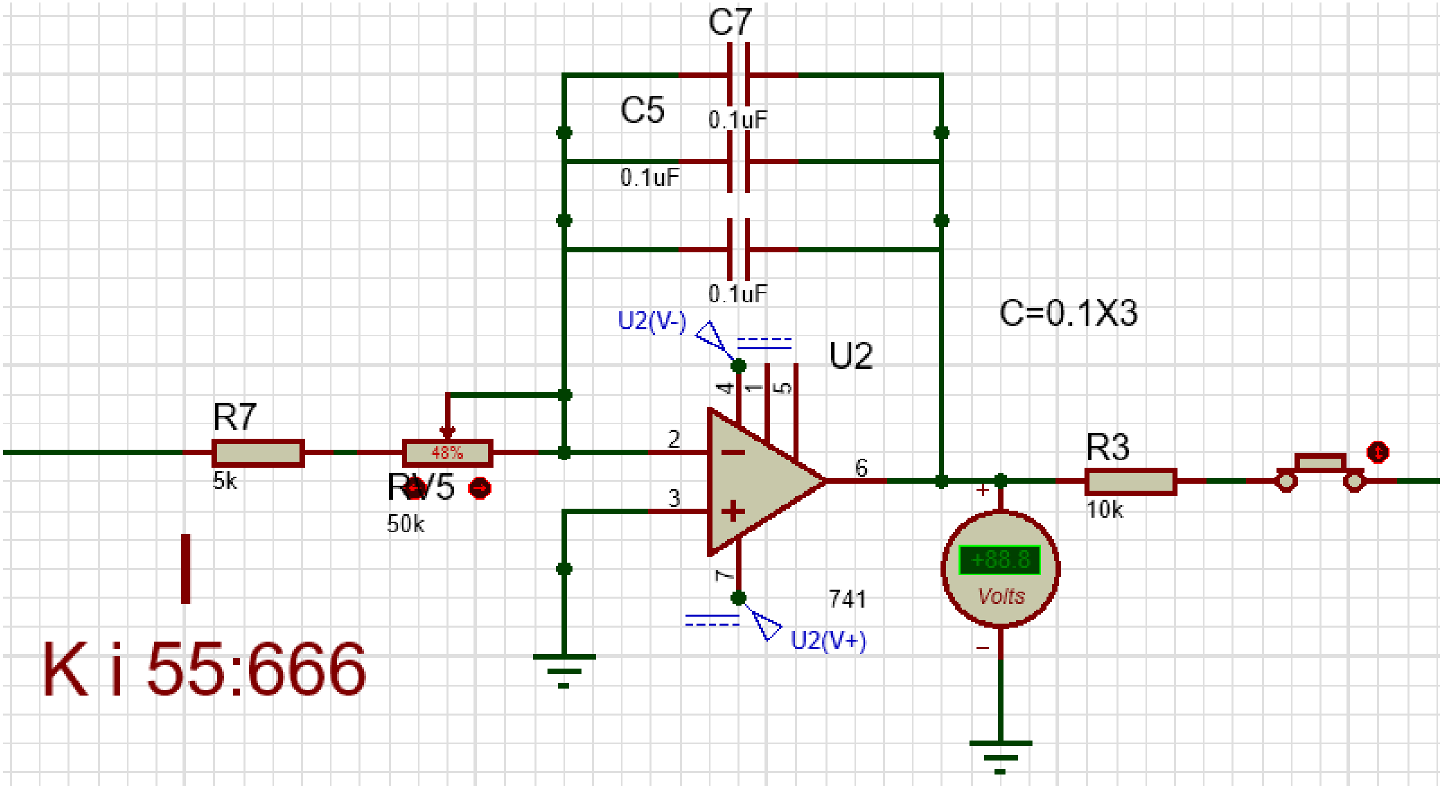
6. **Low-pass filter:**
a filter is added before the circuit to remove any high frequency noise from the system.



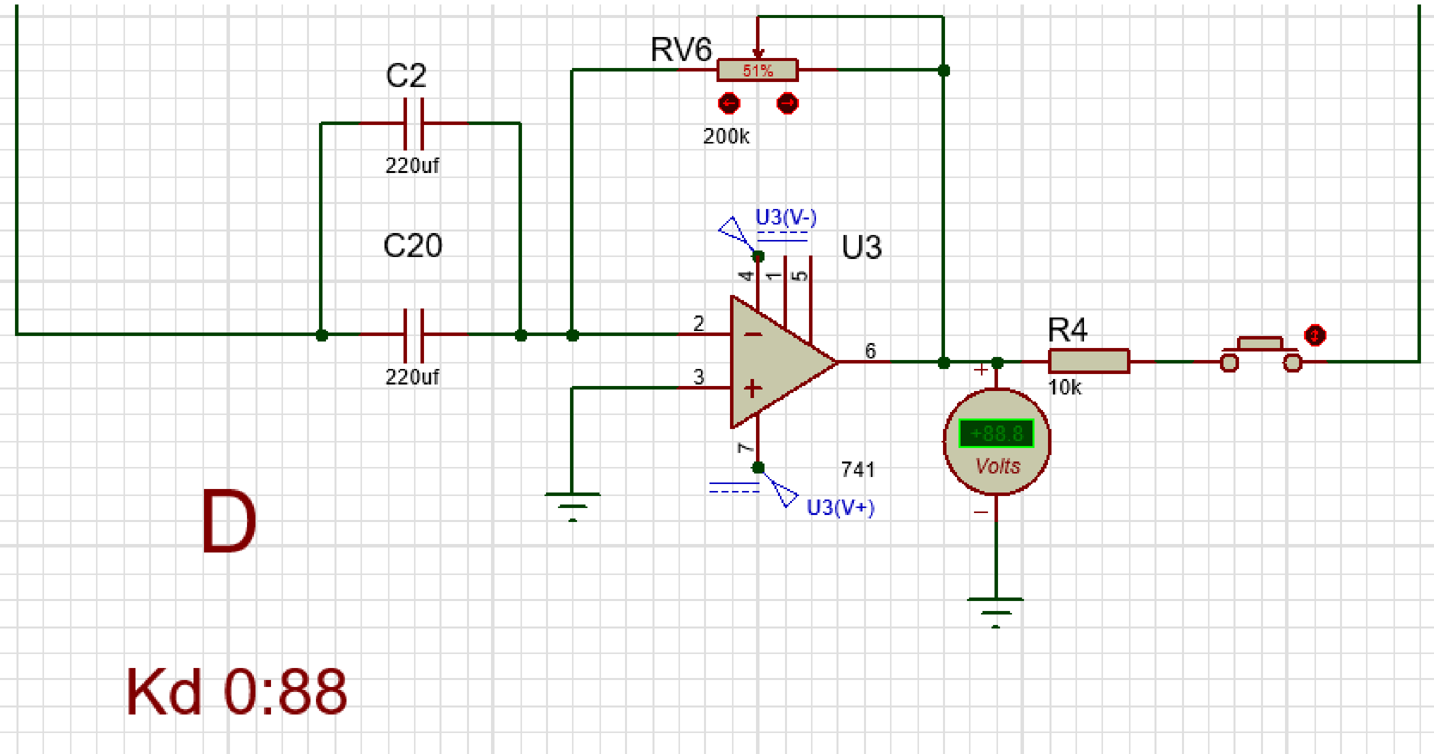
7. **Proportional amplifier:** An inverting amplifier configuration can be used to directly scale the error signal (difference between setpoint and process variable).



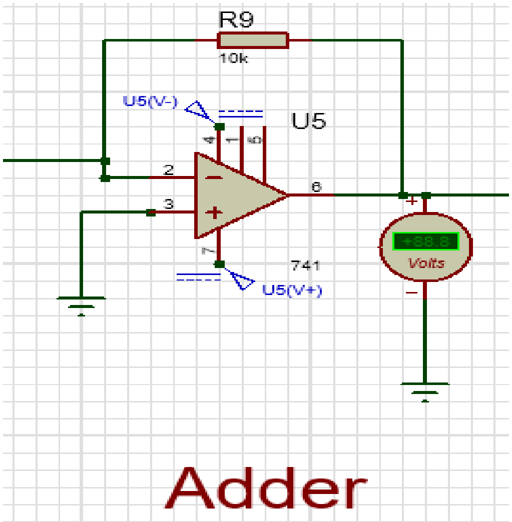
8. **Integrator**: integrator circuit with a capacitor and resistor can accumulate the error signal over time



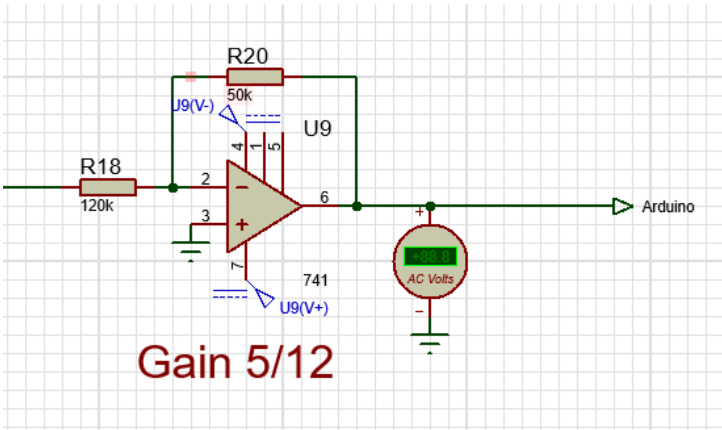
9. **Differentiator**: An op-amp differentiator circuit can provide an output proportional to the rate of change of the error signal.



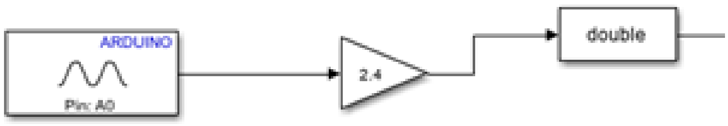
10. Adder: an adder is used to add all three PID signals to one signal that goes into the analog input of the Arduino

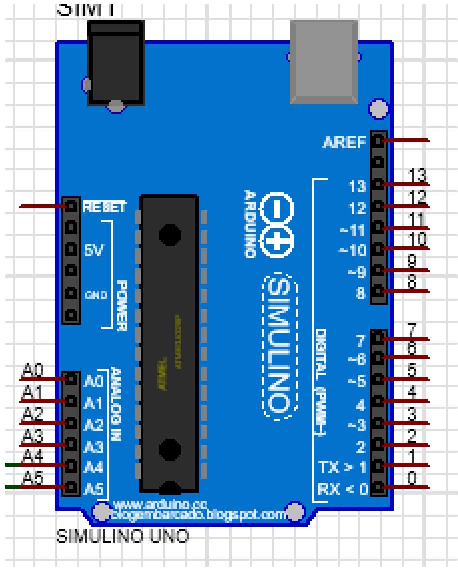


11. Inverting amplifier to scale the voltage to the Arduino 5v



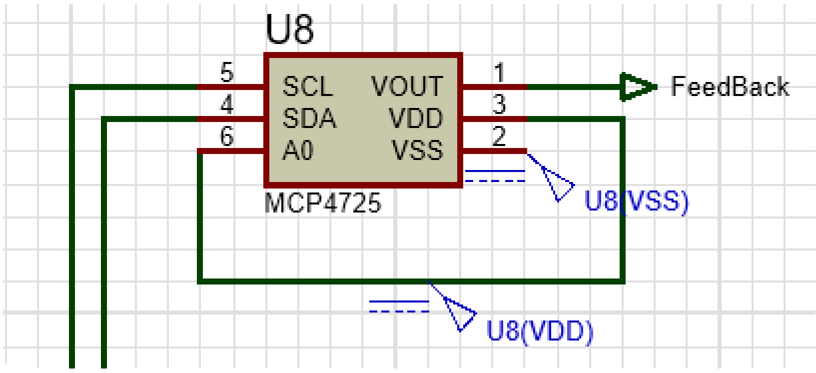
12. Arduino uno: an Arduino is used to interface the circuit with Simulink

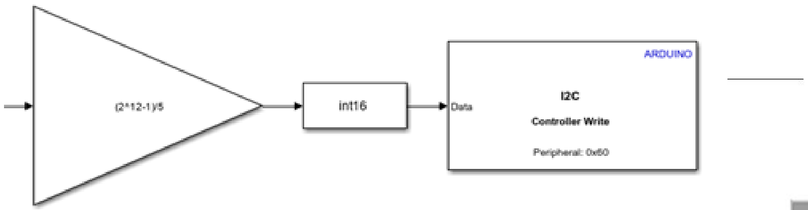




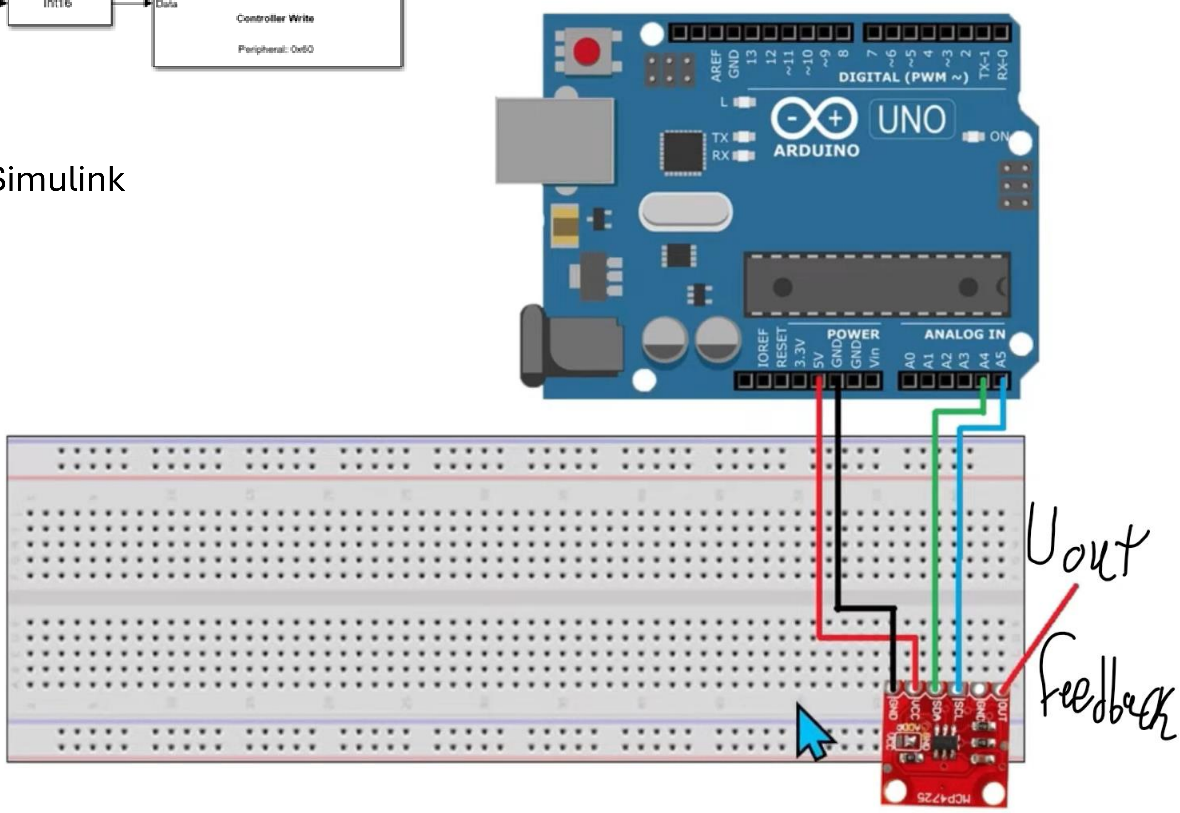
Block Model to input Signal

13. DAC: digital to analog converter is used take the output of the Arduino and convert it to analog feedback signal.





Block Model in Simulink



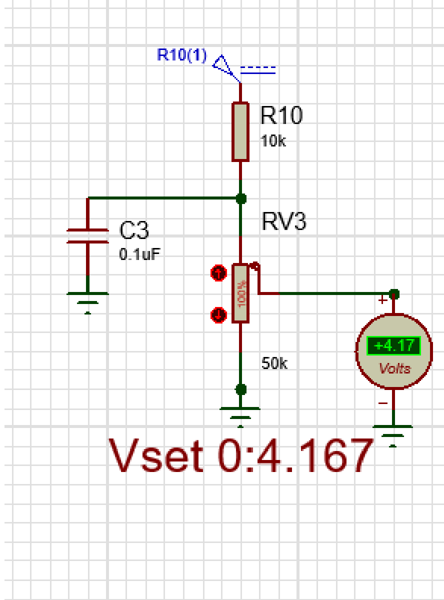
Calculations:

1)Set Point <5v

From this ratio I can reach 0.84v and less

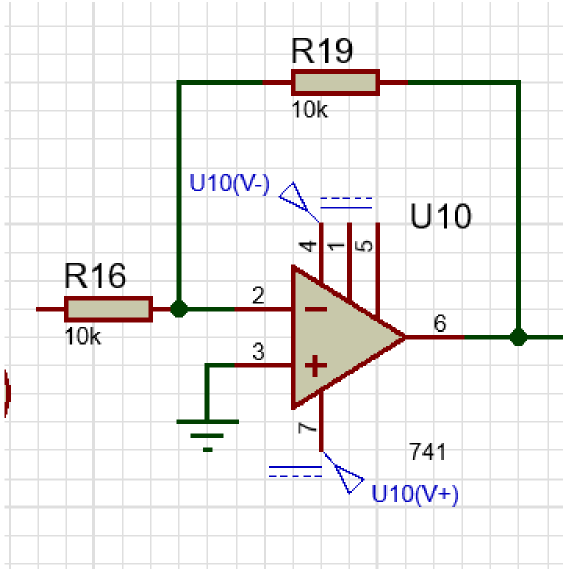
$V(max)=5x\frac{50}{60}$=4.167v

V(min)=0v



2)inverting Amp with Gain=-1Vset

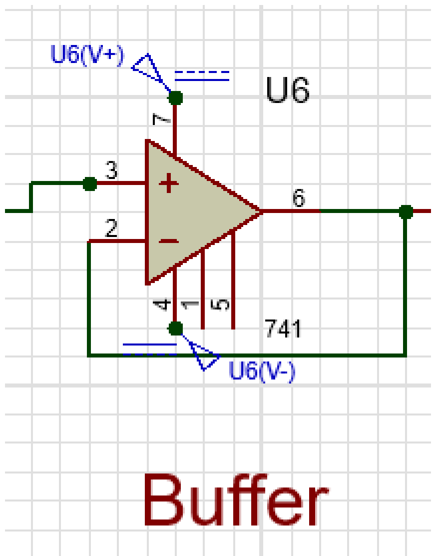
Av=-1



3) buffer

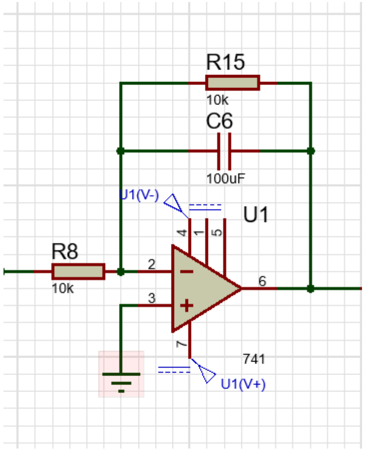
V1=V2

A(v)=1



4)LPF 1:-

To Eliminate Oscillations



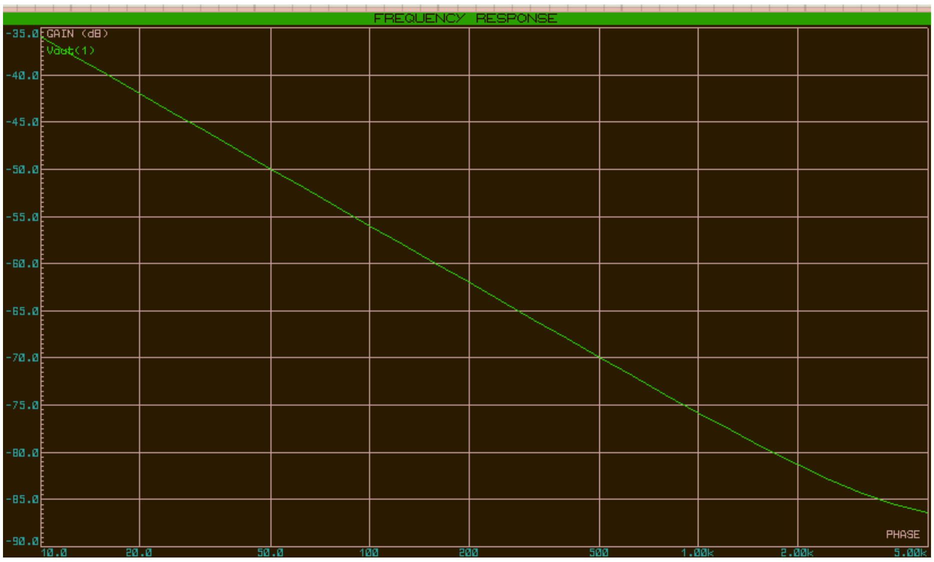
$$A(v) = \frac{-R15/_{R8}}{(C6SR15+1)} = \frac{-1}{S+1}$$

We need RC=1 so I assume c=100uf so R=10k

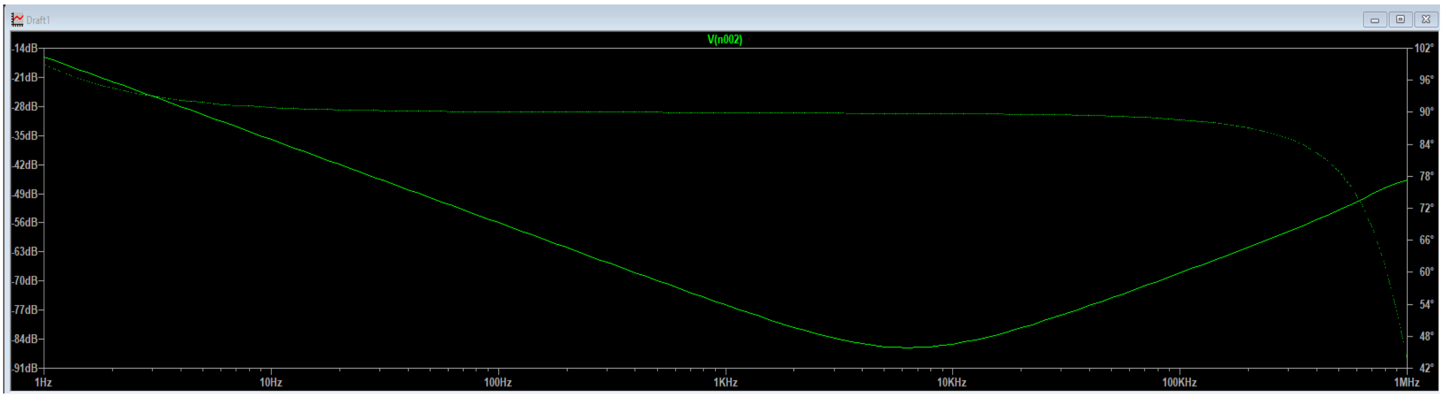
$$H(s) = \frac{1}{S+1}$$

ω=2πf=1 rad/sec

Bode Plot:-



Ac analysis:

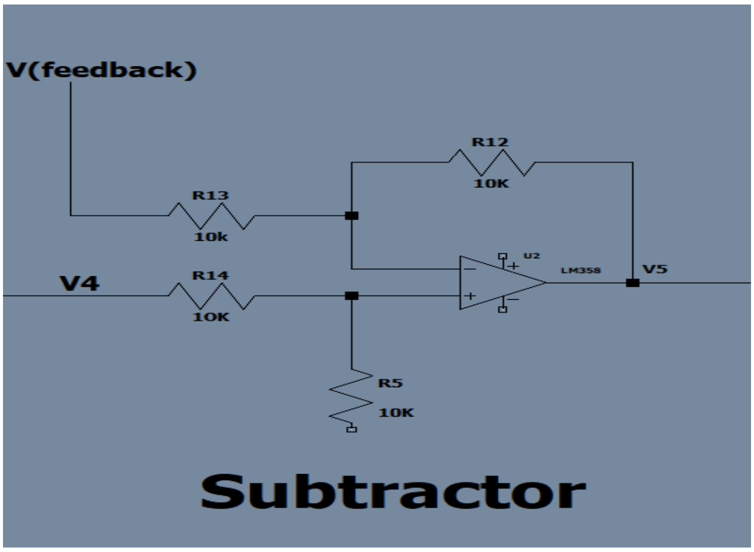


5)subtractor

$$H(s)=\frac{V5}{V4-V(feedback)}=\frac{R12}{R13}$$

Vout=V(feedback)-V(setpoint)

A(v)=1



6)LPF 2:-

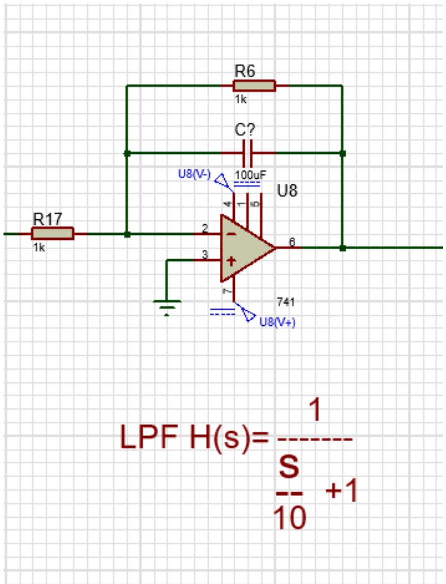
$$H(s)=\frac{V6}{V5}=\frac{1}{CSR6+1}$$

We need RC=0.1 so I assume c=100uf so R=1k

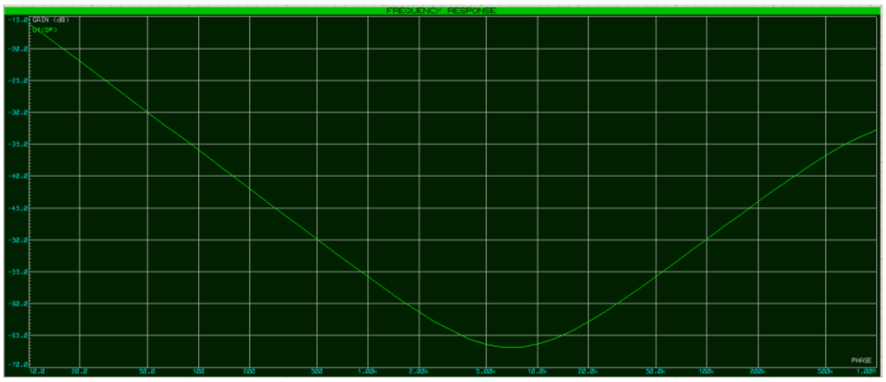
$$H(s)=\frac{1}{\frac{S}{10}+1}$$

To Eliminate Oscillations or noise coming from feedback

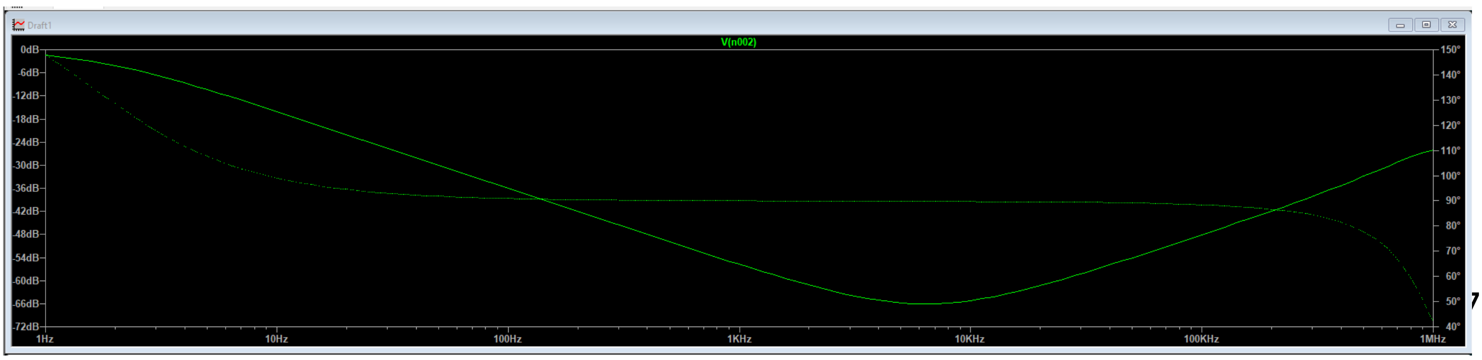
$H(s)=\frac{1}{\frac{s}{10}+1}$, $\omega=2\pi f=10$ rad/sec ,f=1.59 Hz



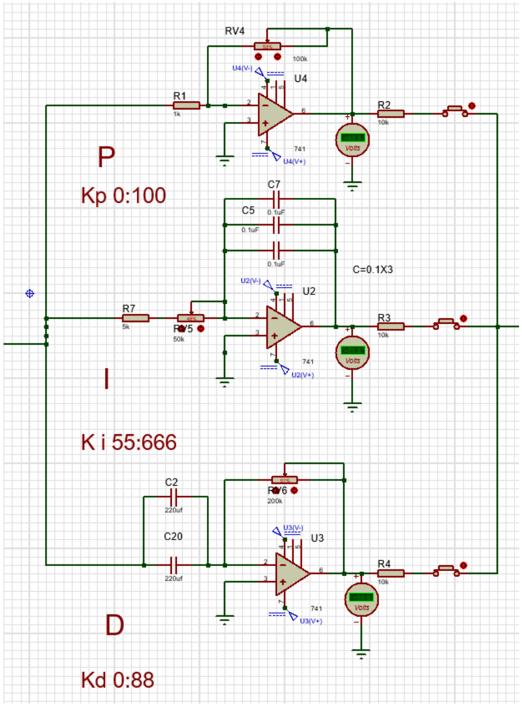
Bode Plot:



Ac analysis:



7)PID



P : A(v)= $\frac{-RV4}{R1}$

Kp=50 when Pot at center →$\frac{50}{1}$=50

I : A(v)= $\frac{-1}{CS(R7+RV5)}$

Since Ki=100 take C=0.3μf and R7+Rv1=(5k+50kpot)=55k

So that the min R=5k and max=105k

We add R=5k to the integrator to to be

far away of infinite gain.

D : A(v)= $-CSRV6$

Since Kd=50 take C=440μf and Rv6=50k (midpoint)

We use 3 Pot in PID To be able to change the values of Kp,I,d.

-As Kp=$\frac{Rpot}{Rconst}$ we use a Pot 100kohm and R1 kohm to vary the Kp in range of

Kp →0:100 .

-As Ki=$\frac{1}{(Rpot+Rconst)C}$ we use a pot and const R to be far away of infinite gain.

we use a Pot 50kohm and R 5kkohm to vary the Ki in range of Ki →55:666 .

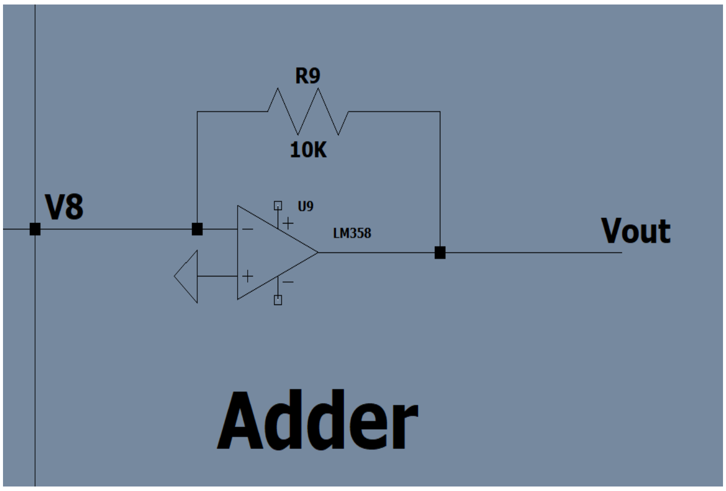
-As Kd=Rpot C we use a pot 50k and 2uf capacitor to vary the Ki in range of

Ki →0:88 .

8)Adder

$$H(s) = \frac{Vout}{V8} = -R9\left(\frac{Vp}{R2} + \frac{Vi}{R3} + \frac{Vd}{R4}\right)$$

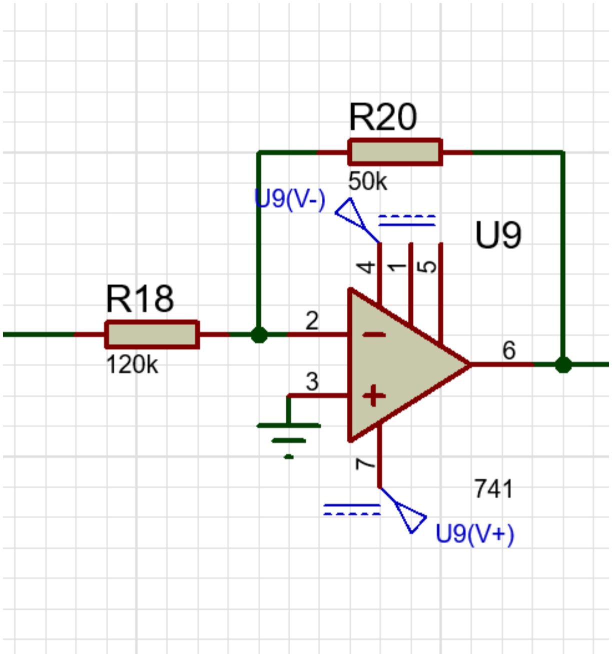
A(v)=-(Vp+Vi+Vd)



*because of we need mapping to send and receive signal to -from arduino we made to op-amp circuit to mappingthe circuit.

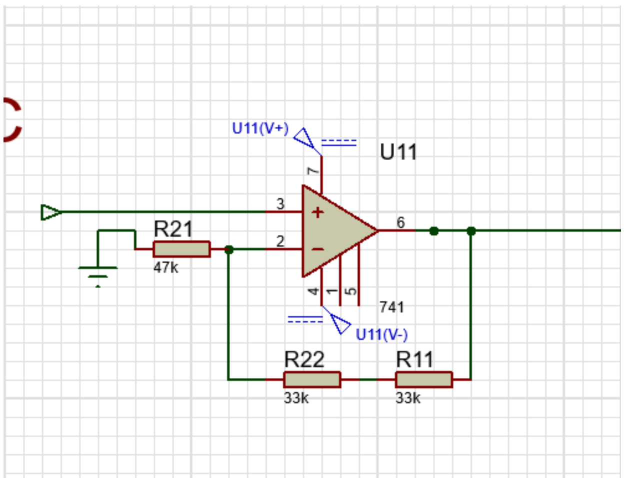
9) inverting amplifier

A(v)= $gain = \frac{-5}{12}$



10) Non-inverting amplifier

A(v)= $gain = 1 + \frac{66}{47} = 12/5$



(P-I-D) Test:-

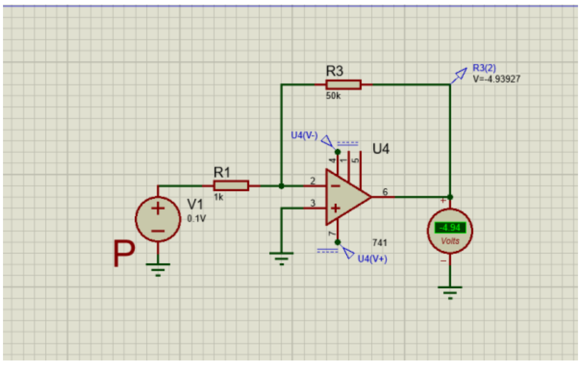
Porpotional Amplifier :- (Kp=50)

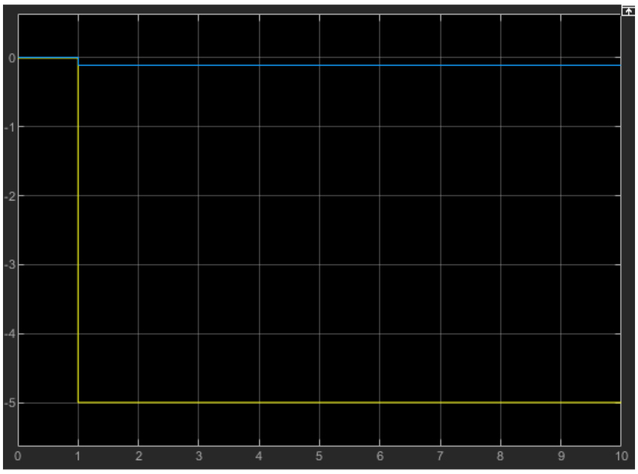
Vin=0.1 v

Vout=- Kp×Vin

Vout(th)=-50×0.1=-5 V

vout(actual)=-5V





Integrator:- (Ki=100)

Vin=0.1 v

Vout= $\frac{\text{Ki}\times\text{Vin}}{\omega}$

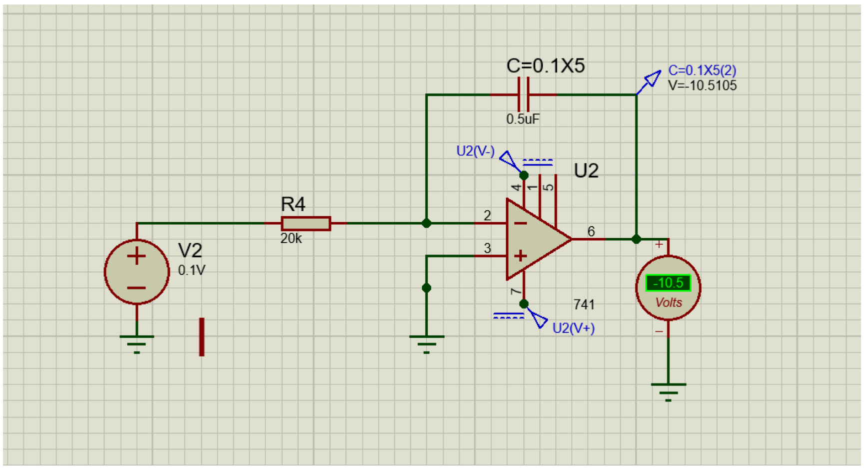
ω=2πf =0

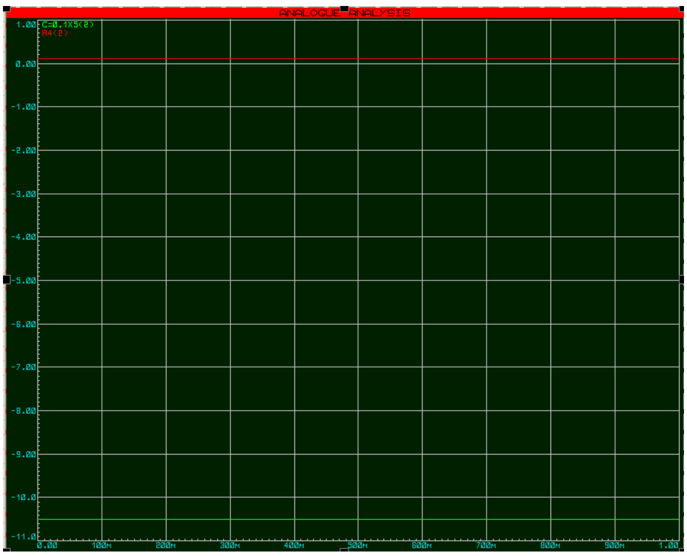
Vout(theoretical) =-infinity

Vout Actual =-11 v

*Dc Voltage will lead to infinite gain

That will be the max volt of o-amp=11v





Differentiator:-(Kd=10)

Vout=- $Kd \times Vin \times \omega$

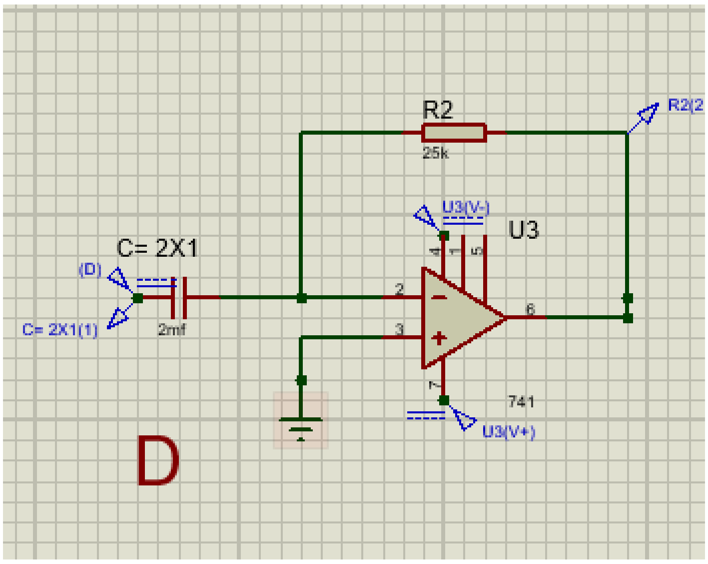
ω=2πf=0

Vin=0.1v

Vout(theoretical)= 0

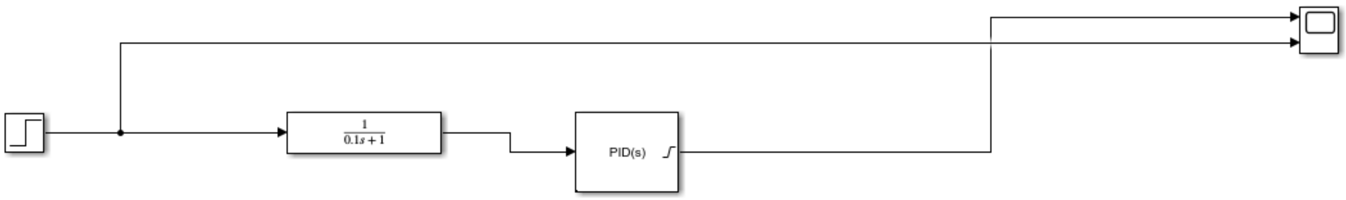
Vout Actual =0.01mv

*Dc voltage will lead to 0 gain.

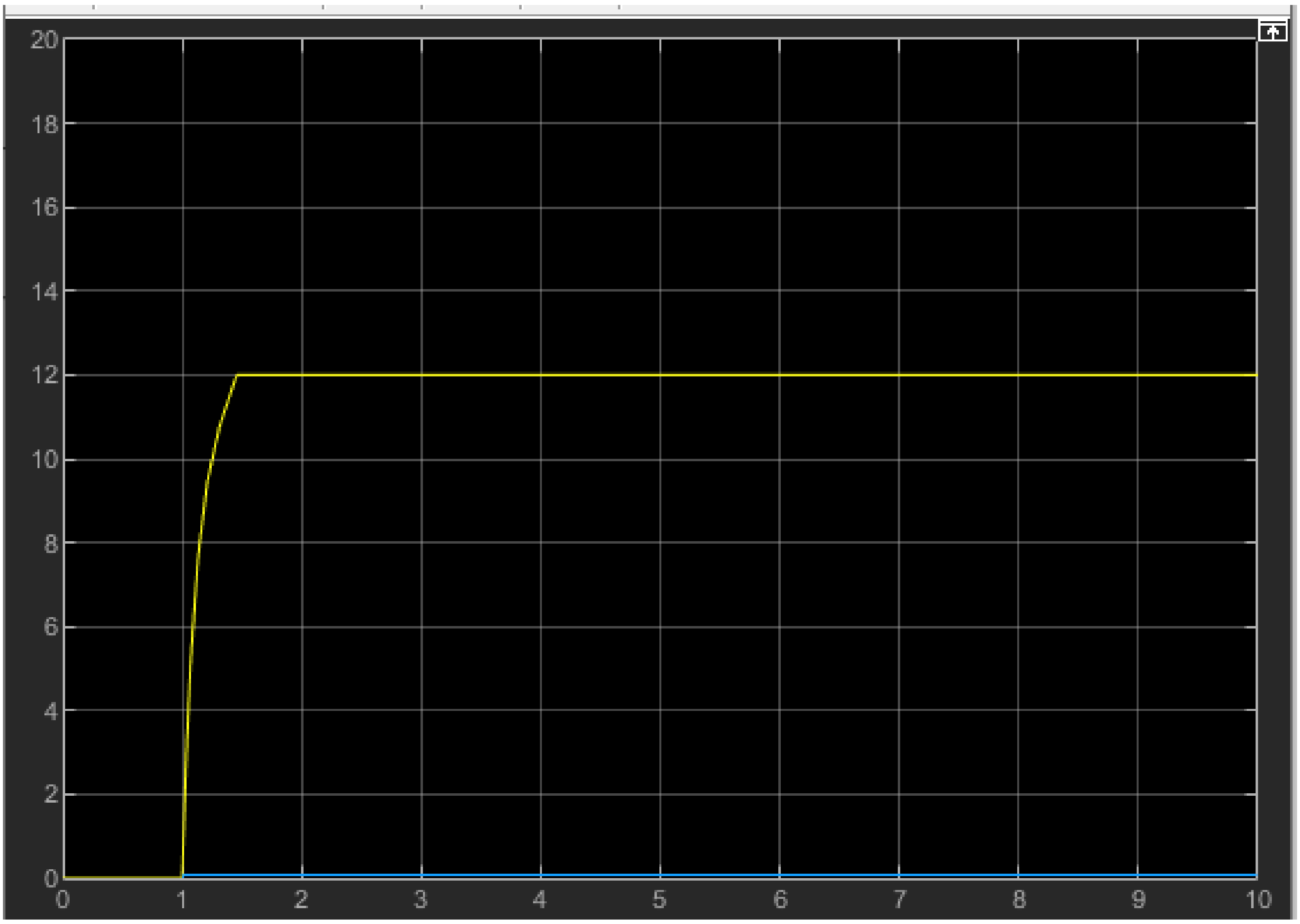


PID Test:-

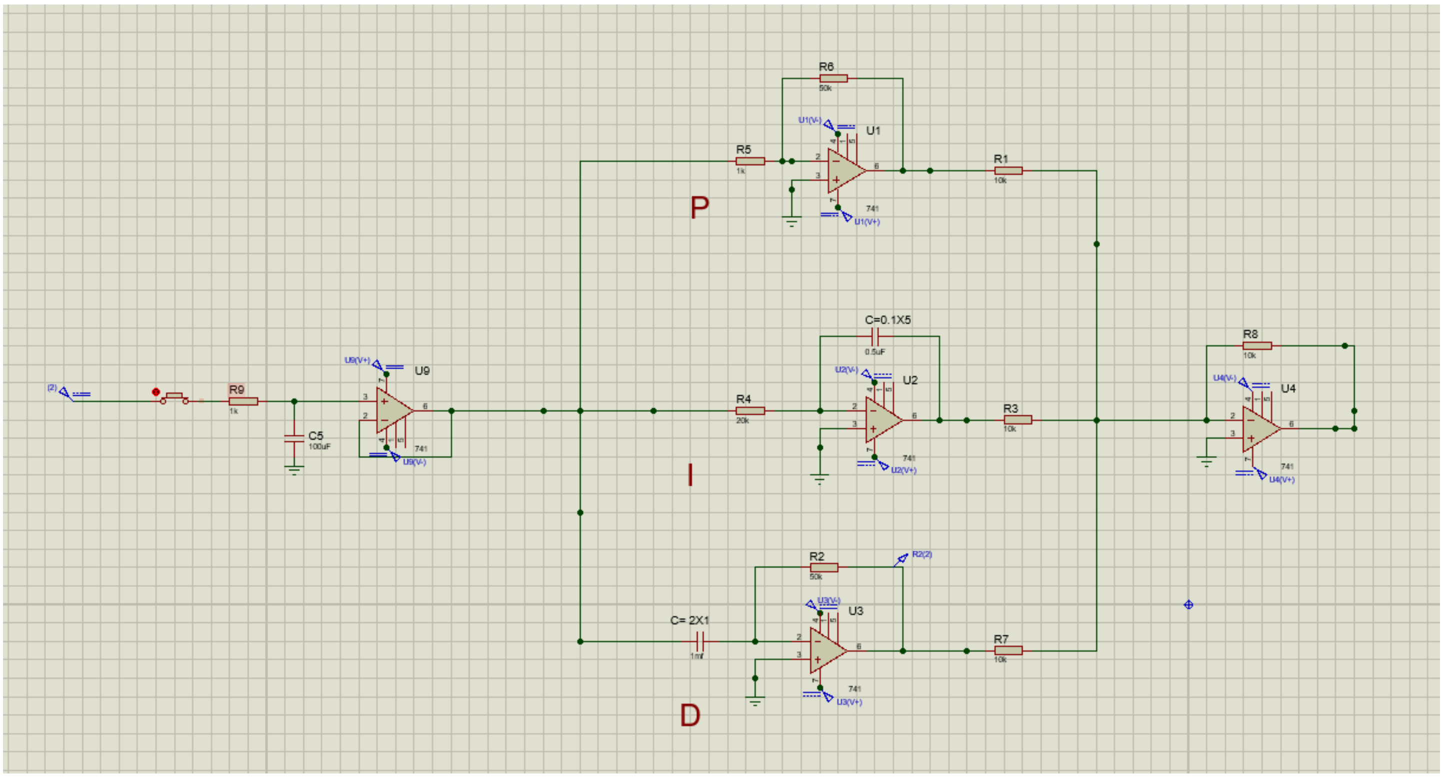
1)Simulated in Simulink:-



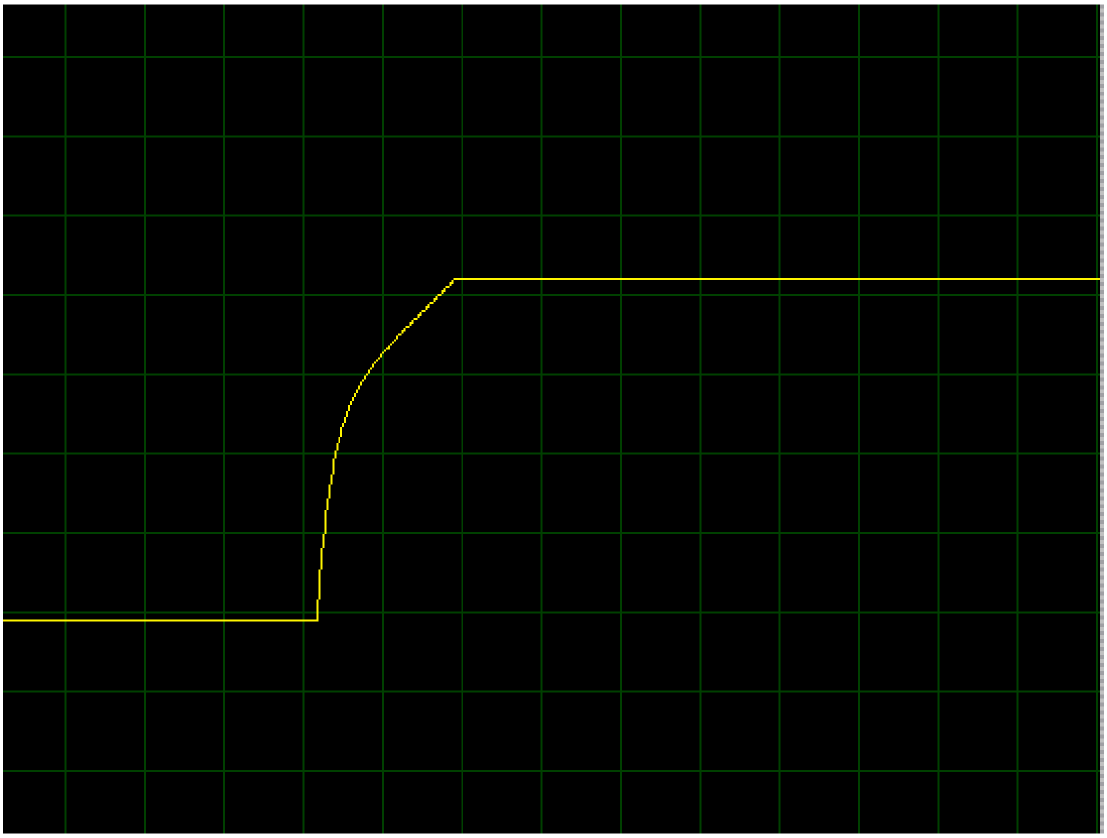
As Vin=0.1v(DC) the Vout is the saturated volt of the PID Block .



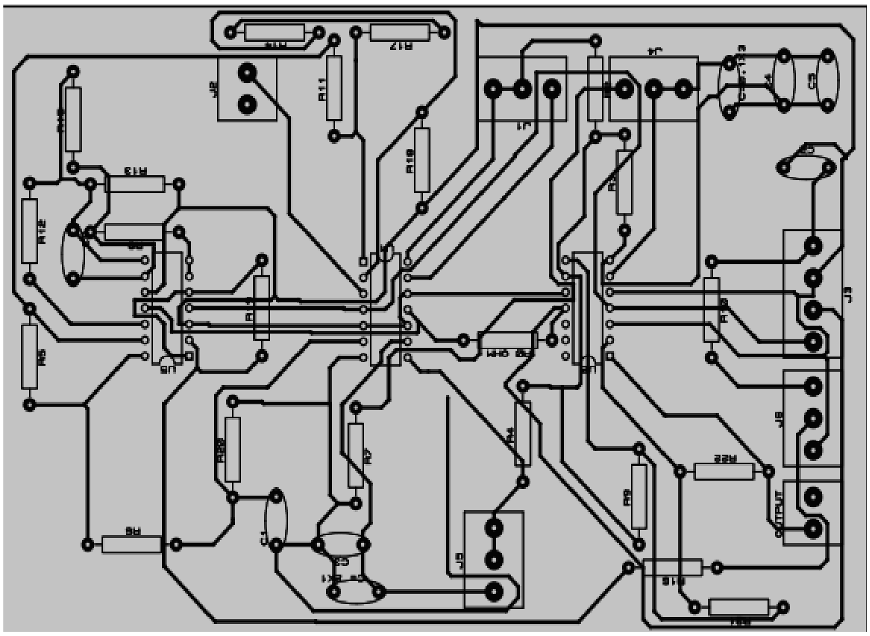
2)Simulated in Proteus:-

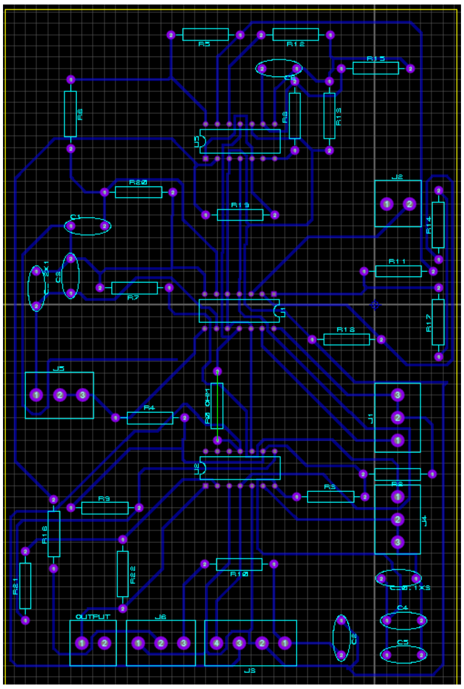


As Vin=0.1v(DC) the Vout is the max volt of the OP-Amp .



PCB Layout:-

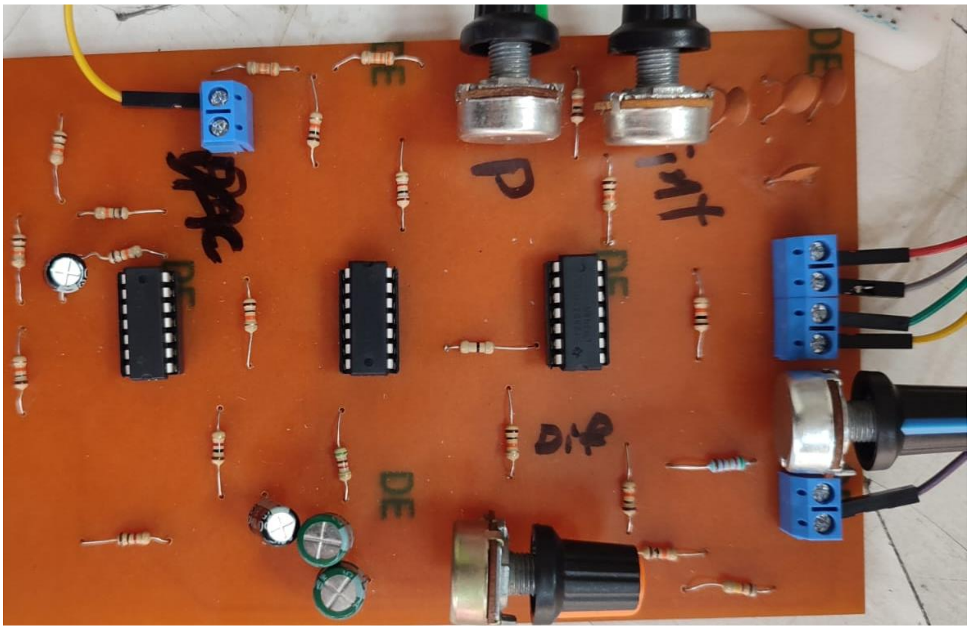


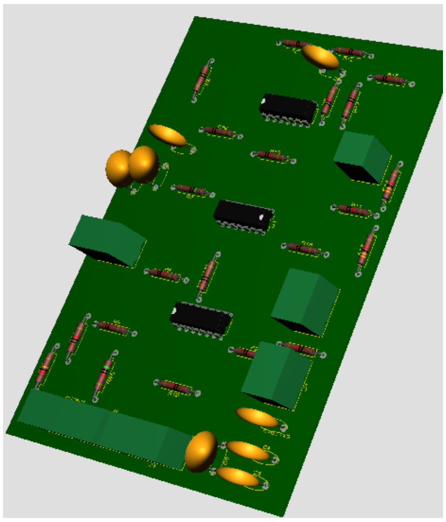


PCB Implementation: -

We use OP-Amp LM348N because we need the PCB to be compact as possible.

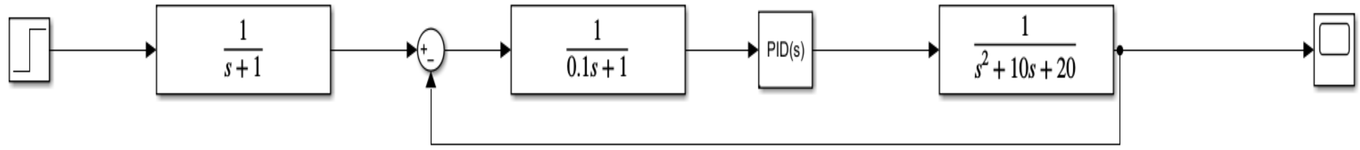
The ic has 4 op-amp .

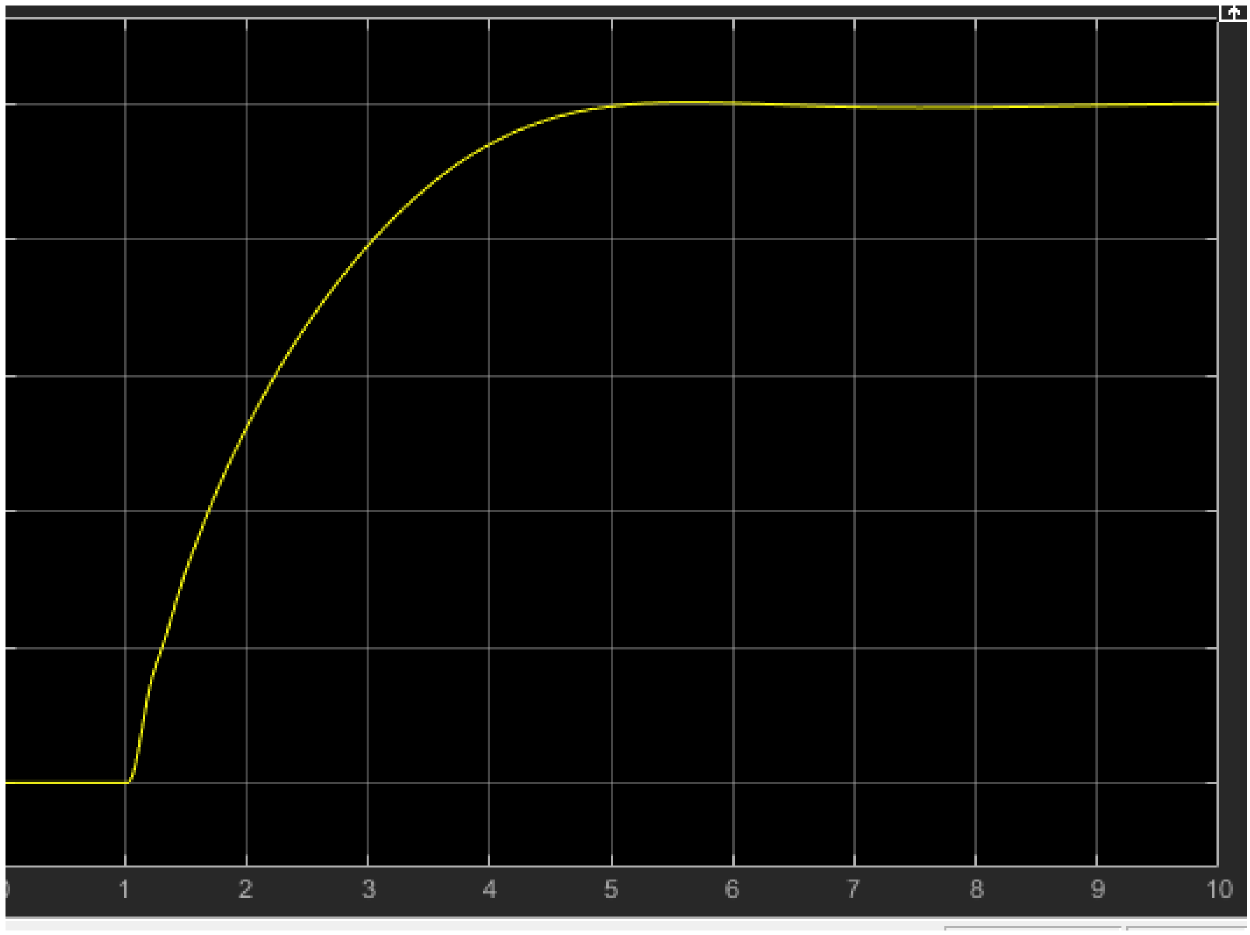




All System Simulated in Simulink:-

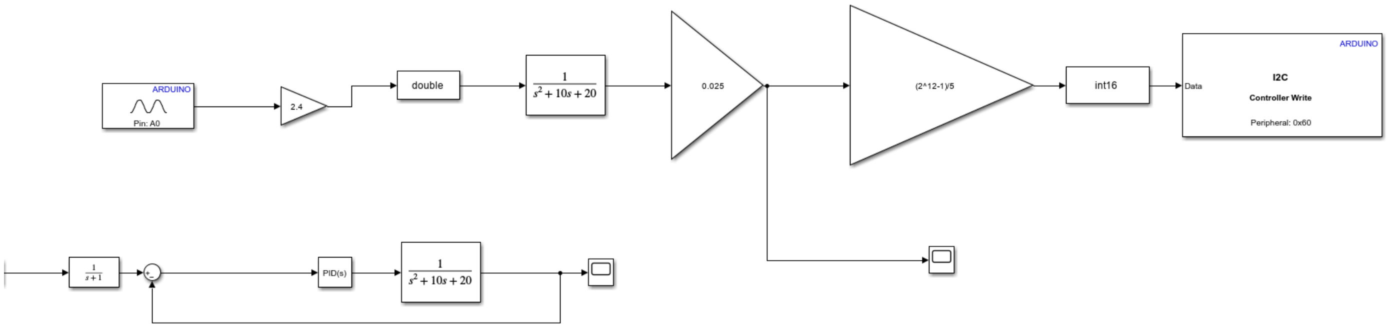
With Vin=0.1





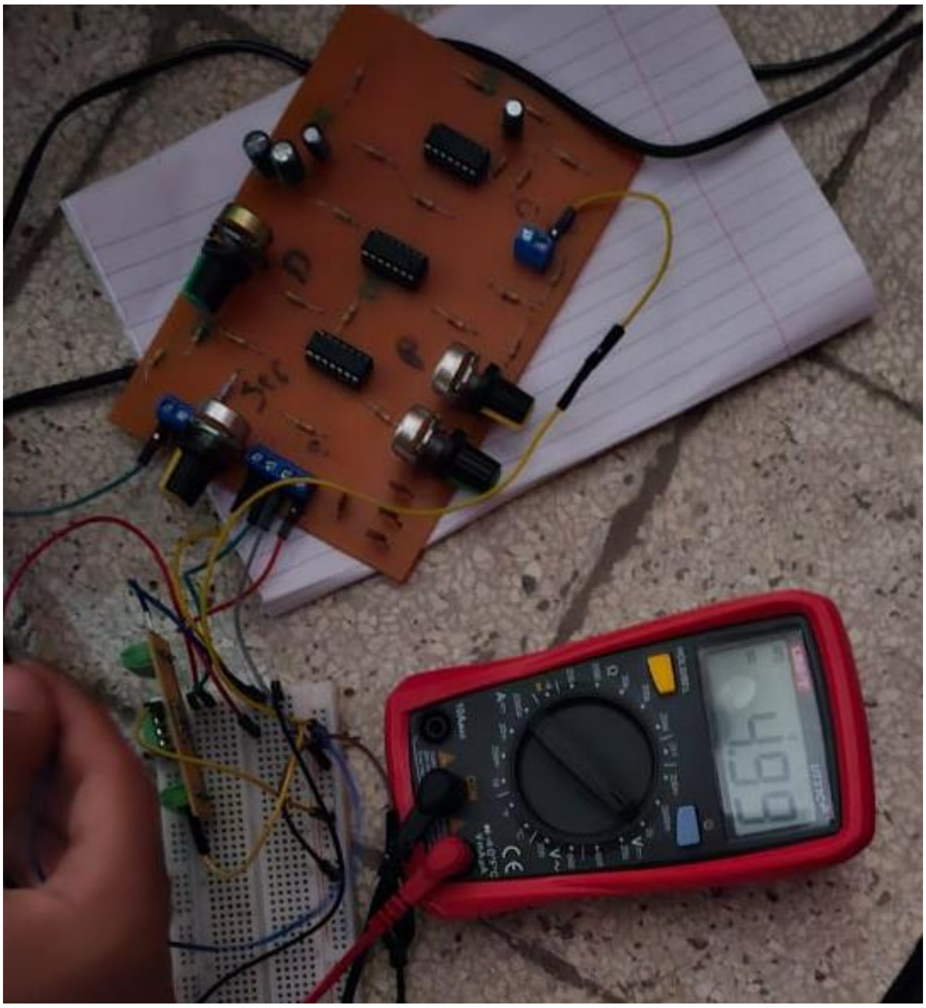
Implementation the project with Simulink :-

1)Block diagram in simulink :



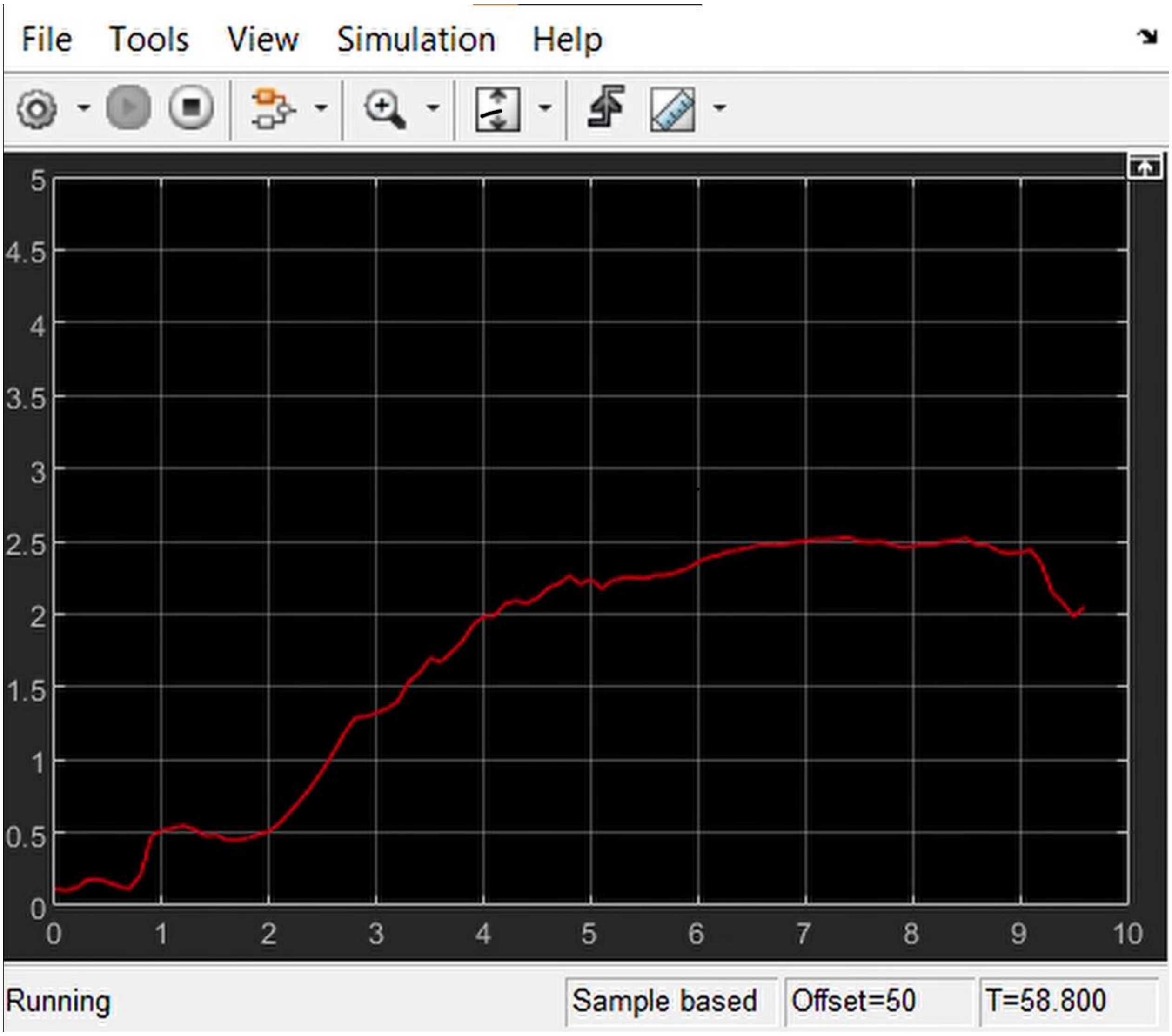
2)Test the Pid Response :-

vout=4.99V from P-I-D with feedback =0



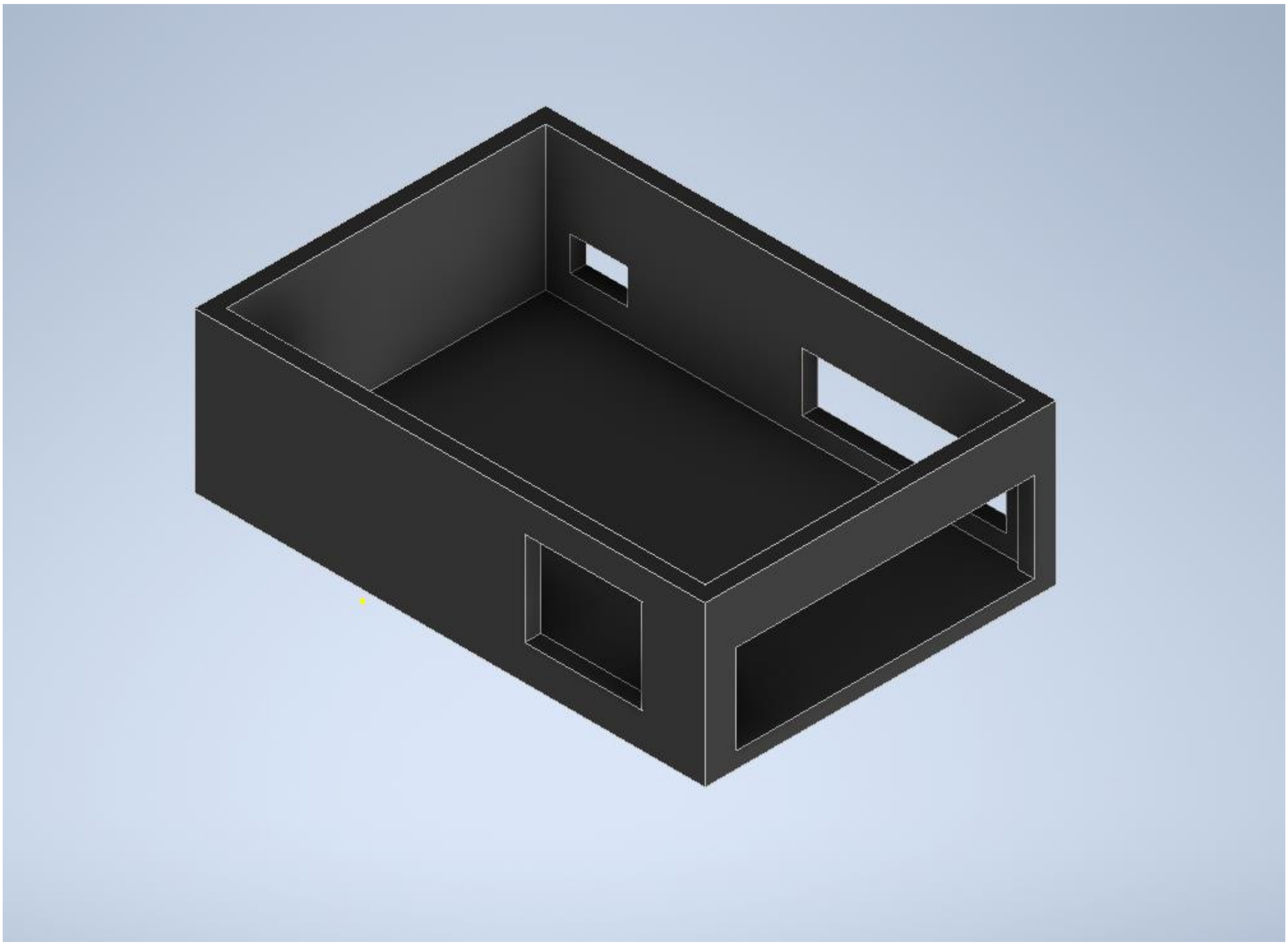
TF Response:-

Set point =2.51v the Transfer Function response varies about 2.5v



System Encloser:

Not implemented



Conclusion: -

The simulated curve of implemented circuit has many Errors it should be

1- From the Simulink Block model
2- the communication between the DAC and Arduino and PID has something Wrong.

*We make sure that all connection are good and all op-amp work good

So we sure that the Error From Simulink.